

HTTP Lab

The image shows a Wireshark packet capture window titled "*Wi-Fi". The interface includes a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help) and a toolbar. A filter bar at the top shows "http". The packet list pane displays four packets:

No.	Time	Source	Destination	Protocol	Length	Info
753	2.056947800	10.0.0.86	128.119.245.12	HTTP	546	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
763	2.110797400	128.119.245.12	10.0.0.86	HTTP	543	HTTP/1.1 200 OK (text/html)
769	2.175963700	10.0.0.86	128.119.245.12	HTTP	492	GET /favicon.ico HTTP/1.1
812	2.231498000	128.119.245.12	10.0.0.86	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)

The packet details pane shows the selected packet (No. 753) expanded to show the Hypertext Transfer Protocol section:

- Transmission Control Protocol, Src Port: 61545, Dst Port: 80, Seq: 1,
- Hypertext Transfer Protocol
 - GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /wireshark-labs/HTTP-wireshark-file1.html
 - Request Version: HTTP/1.1
 - Host: gaia.cs.umass.edu\r\n
 - Connection: keep-alive\r\n
 - Upgrade-Insecure-Requests: 1\r\n
 - User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/5
 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image
 - Accept-Encoding: gzip, deflate\r\n
 - Accept-Language: en-US,en;q=0.9\r\n
 - dnt: 1\r\n
 - sec-gpc: 1\r\n
 - \r\n

The packet bytes pane shows the raw data of the selected packet, with the first few bytes highlighted in red: 94 04 e3 a1 8f 13 48 45 e6 8b f6 df 08 00 45 00.

At the bottom, the status bar indicates: Destination Hardware Address (eth.dst), 6 bytes. Packets: 930 · Displayed: 4 (0.4%) · Dropped: 0 (0.0%) Profile: Default.

1. Browser: HTTP/1.1 | Server: HTTP/1.1
2. Accept-Language: en-US,en;q=0.9
3. My PC: 10.0.0.86 && gaia: 128.119.245.12

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The image shows a Wireshark packet capture window titled "*Wi-Fi". The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. The packet list pane shows four packets, with packet 763 selected. The packet details pane shows the structure of the selected packet, and the packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
753	2.056947800	10.0.0.86	128.119.245.12	HTTP	546	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
763	2.110797400	128.119.245.12	10.0.0.86	HTTP	543	HTTP/1.1 200 OK (text/html)
769	2.175963700	10.0.0.86	128.119.245.12	HTTP	492	GET /favicon.ico HTTP/1.1
812	2.231498000	128.119.245.12	10.0.0.86	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)

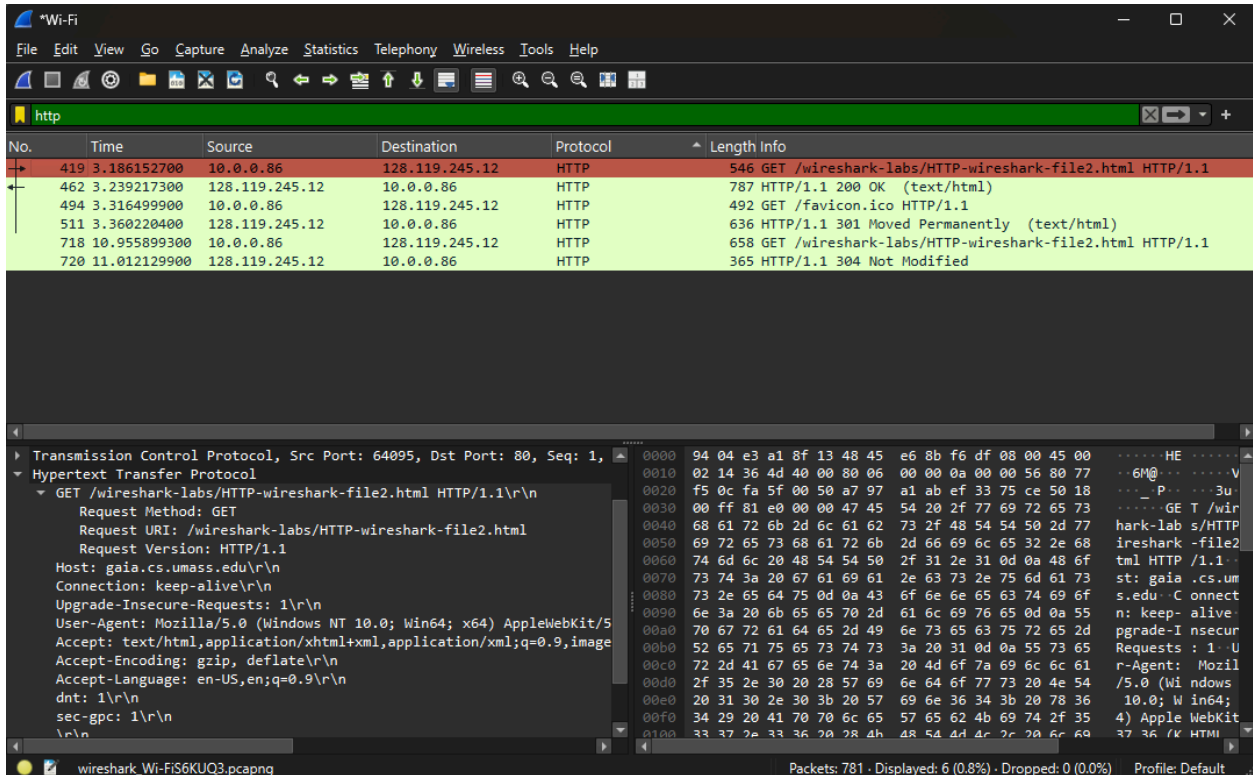
Packet 763 details:

- Transmission Control Protocol, Src Port: 80, Dst Port: 61545, Seq: 1, Win: 0, Len: 0
- Hypertext Transfer Protocol
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Date: Sun, 14 Jun 2026 20:07:17 GMT\r\n
 - Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.5 mod_fcgid/2.3.9 mod_ssl/3.5.5
 - Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
 - ETag: "80-64231b6717e87"\r\n
 - Accept-Ranges: bytes\r\n
 - Content-Length: 128\r\n
 - Keep-Alive: timeout=5, max=100\r\n
 - Connection: Keep-Alive\r\n
 - Content-Type: text/html; charset=UTF-8\r\n

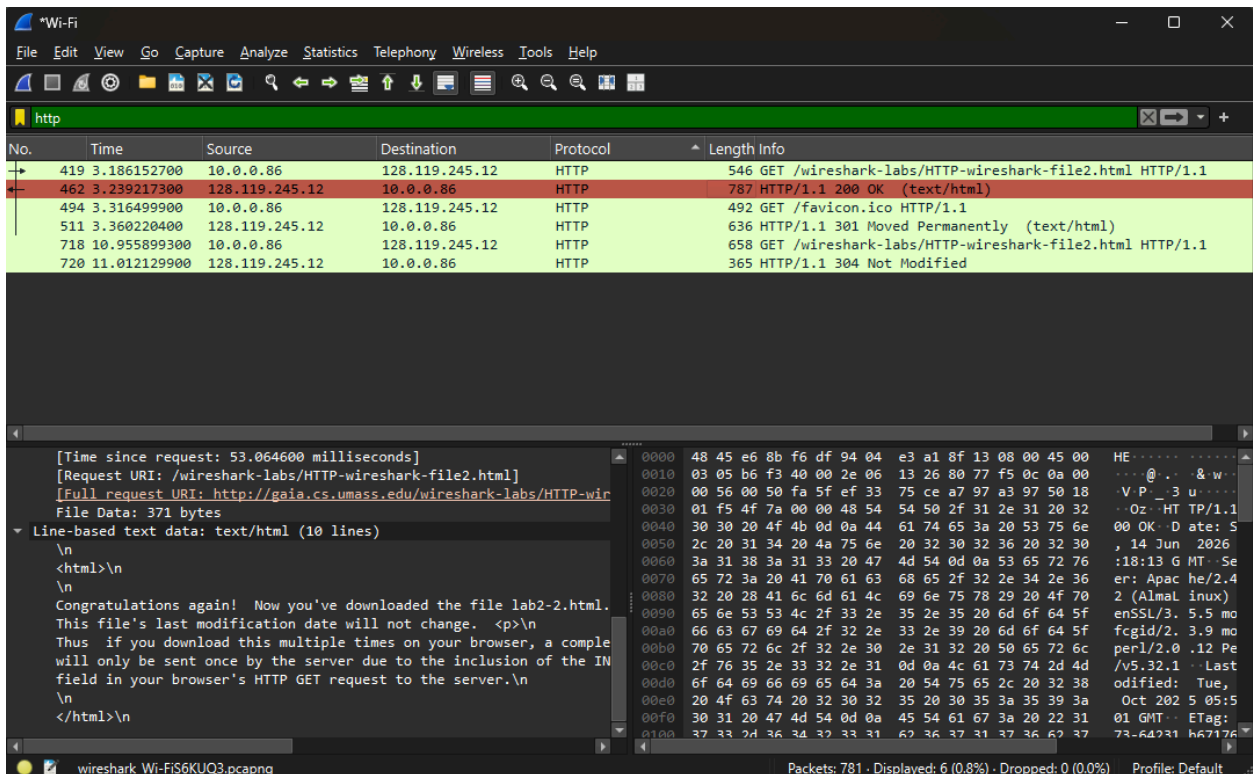
Packet bytes pane shows the raw data of the packet, including the status code 200 and the response phrase OK.

- Status code 200 OK, server running HTTP/1.1
- The HTML file was last modified on Tuesday, October 28 2025 at 05:59:01 GMT
- 128 bytes of content are being returned to my browser.
- No, there are no additional headers visible in the raw data that aren't already displayed in the packet details panel. All headers are accounted for.

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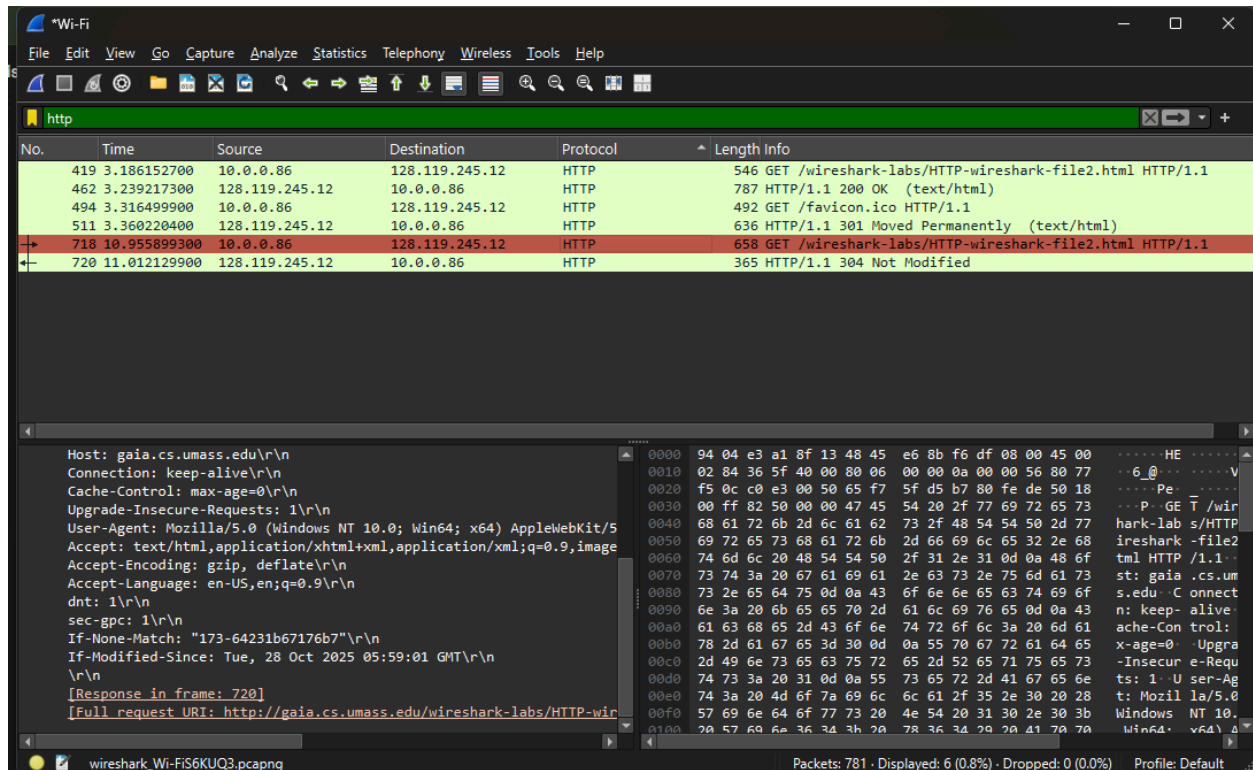


8. No, there is no If-Modified-Since line in the first HTTP GET. This is because the browser had no cached version of the file (we cleared the cache), so it had no date to send.



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9. Yes, the server explicitly returned the contents of the file. We can tell because the response has status code 200 OK and contains the full HTML body (371 bytes of File Data with visible <html> content).



10. Yes, the second GET does contain an If-Modified-Since: header. It says: Tue, 28 Oct 2025 05:59:01 GMT. This is the same date from the first response's Last-Modified field.
11. The server returned status code 304 Not Modified. The server did not explicitly return the contents of the file, no HTML body was sent. This is because the file hadn't changed since Tue, 28 Oct 2025 05:59:01 GMT, so the server just told the browser to use its cached copy instead of sending the file again.

HTTP Lab

The image shows a Wireshark capture of HTTP traffic. The top pane displays a list of packets with the following details:

No.	Time	Source	Destination	Protocol	Length	Info
302	3.196079100	10.0.0.86	128.119.245.12	HTTP	546	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
338	3.241178100	128.119.245.12	10.0.0.86	HTTP	538	HTTP/1.1 200 OK (text/html)
373	3.324557300	10.0.0.86	128.119.245.12	HTTP	492	GET /favicon.ico HTTP/1.1
389	3.370352300	128.119.245.12	10.0.0.86	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)

The bottom pane shows the details of the selected packet (Frame 302). The details are as follows:

- Frame 302: Packet, 546 bytes on wire (4368 bits), 546 bytes captured
- Ethernet II, Src: CloudNetwork_8b:f6:df (48:45:e6:8b:f6:df), Dst: Vant
- Internet Protocol Version 4, Src: 10.0.0.86, Dst: 128.119.245.12
- Transmission Control Protocol, Src Port: 58999, Dst Port: 80, Seq: 1,
- Hypertext Transfer Protocol
 - GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /wireshark-labs/HTTP-wireshark-file3.html
 - Request Version: HTTP/1.1
 - Host: gaia.cs.umass.edu\r\n
 - Connection: keep-alive\r\n
 - Upgrade-Insecure-Requests: 1\r\n
 - User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/5
 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image
 - Accept-Encoding: gzip, deflate\r\n
 - Accept-Language: en-US,en;q=0.9\r\n

The status bar at the bottom indicates: Packets: 565 · Displayed: 4 (0.7%) · Dropped: 0 (0.0%) Profile: Default

12. Browser sent 1 HTTP GET request. It is in packet number **302**.
13. The status code and phrase are contained in packet number **338**.
14. Status code 200 OK
15. 4 TCP segments were needed to carry the HTTP response and the Bill of Rights text:
 - a. Segment 1: packet 335 (1460 bytes)
 - b. Segment 2: packet 336 (1460 bytes)
 - c. Segment 3: packet 337 (1460 bytes)
 - d. Segment 4: packet 338 (484 bytes)
 - i. Total: 4864 bytes

HTTP Lab

The image shows a Wireshark capture of network traffic. The top pane displays a list of captured packets, with the first four packets highlighted in green. The second pane shows the details of the selected packet (No. 662), including the Ethernet II header, Internet Protocol Version 4 header, and the Hypertext Transfer Protocol (HTTP) header. The third pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
662	3.398142500	10.0.0.86	10.0.0.243	HTTP	226	GET /dial/dd.xml HTTP/1.1
668	3.468830700	10.0.0.243	10.0.0.86	HTTP/XML	1360	HTTP/1.1 200 OK
672	3.480227700	10.0.0.86	10.0.0.243	HTTP	244	GET /dial/com.spotify.Spotify.TVv2 HTTP/1.1
678	3.499715800	10.0.0.243	10.0.0.86	HTTP	141	HTTP/1.1 404 Not Found

Frame 662: Packet, 226 bytes on wire (1808 bits), 226 bytes captured (1808 bits) on interface 0
Ethernet II, Src: CloudNetwork 8b:f6:df (48:45:e6:8b:f6:df), Dst: Huawei 10:00:00:00:00:00
Internet Protocol Version 4, Src: 10.0.0.86, Dst: 10.0.0.243
Hypertext Transfer Protocol
GET /dial/dd.xml HTTP/1.1
User-Agent: Spotify/129.200147.0
Host: 10.0.0.243:8066
Keep-Alive: 0
Accept-Encoding: gzip
Connection: keep-alive

16. The browser sent 2 HTTP GET requests, one to `gaia.cs.umass.edu` for the base HTML and publisher logo, and one to `caite.cs.umass.edu` for the book cover image. The 404 on one of them means that server no longer has the image.
17. The two images were downloaded in parallel. We can tell because the second GET was sent at 3.480 seconds, only ~82ms after the first GET at 3.398 seconds, the browser didn't wait for the first response before sending the second request.

HTTP Lab

The image shows a Wireshark network traffic capture. The top pane displays a list of captured packets. The second pane shows the details of the selected packet (No. 323), which is an HTTP 401 Unauthorized response. The third pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
143	1.880292800	10.0.0.86	128.119.245.12	HTTP	562	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.ht
178	1.938351200	128.119.245.12	10.0.0.86	HTTP	774	HTTP/1.1 401 Unauthorized (text/html)
323	12.019063600	10.0.0.86	128.119.245.12	HTTP	647	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.ht
326	12.075063200	128.119.245.12	10.0.0.86	HTTP	547	HTTP/1.1 200 OK (text/html)
342	12.153287300	10.0.0.86	128.119.245.12	HTTP	508	GET /favicon.ico HTTP/1.1
344	12.198237200	128.119.245.12	10.0.0.86	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
568	19.042243900	10.0.0.86	10.0.0.243	HTTP	226	GET /dial/dd.xml HTTP/1.1

Details of packet 323 (HTTP 401 Unauthorized):

- Request Method: GET
- Request URI: /wireshark-labs/protected_pages/HTTP-wireshark-file5.html
- Request Version: HTTP/1.1
- Host: gaia.cs.umass.edu
- Connection: keep-alive
- Cache-Control: max-age=0
- Authorization: Basic d2lyZXNoYXJrLXN0dWRlbnRzOm5ldHdvcmVs=
- Upgrade-Insecure-Requests: 1
- User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2703.104 Safari/537.36

Raw packet data (hex and ASCII):

```
00c0 78 2d 61 67 65 3d 30 0d 0a 41 75 74 68 6f 72 69 x-age=0-Autho
00d0 7a 61 74 69 6f 6e 3a 20 42 61 73 69 63 20 64 32 zation: Basic
00e0 6c 79 5a 58 4e 6f 59 58 4a 72 4c 58 4e 30 64 57 lyZXNoYXJrLXN0
00f0 52 6c 62 6e 52 7a 4f 6d 35 6c 64 48 64 76 63 6d RlbnRzOm5ldHdv
0100 73 3d 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 s=..Upgr ade-In
0110 63 75 72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 cure-Req uests:
0120 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f ..User-A gent:
0130 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f zilla/5.0 (Win
0140 77 73 20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 ws NT 10.0; Wi
0150 34 3b 20 78 36 34 29 20 41 70 70 6c 65 57 65 62 4; x64) AppleW
0160 4b 69 74 2f 35 33 37 2e 33 36 20 28 4b 48 54 4d Kit/537.36 (KH
0170 4c 2c 20 6c 69 6b 65 20 47 65 63 6b 6f 29 20 43 L, like Gecko)
0180 68 72 6f 6d 65 2f 31 34 39 2e 30 2e 30 2e 30 20 hrome/14 9.0.0.
0190 53 61 66 61 72 69 2f 35 33 37 2e 33 36 0d 0a 41 Safari/5 37.36
01a0 63 63 65 70 74 3a 20 74 65 78 74 2f 68 74 6d 6c ccept: text/ht
01b0 2c 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 68 74 applica tion/v
```

18. The server returned 401 Unauthorized in response to the initial GET. This is the server telling the browser you need to provide credentials to access this page.
19. The second GET includes a new Authorization: Basic header with the value d2lyZXNoYXJrLXN0dWRlbnRzOm5ldHdvcmVs=, this is the username wireshark-students and password network encoded in Base64 format.